

PORTFOLIO PROJECT

Online Retail Sales Analysis

End-to-end commercial analytics and Power BI reporting

A reproducible sales analytics workflow that transforms transaction data into revenue, customer, product and country-level business insight.

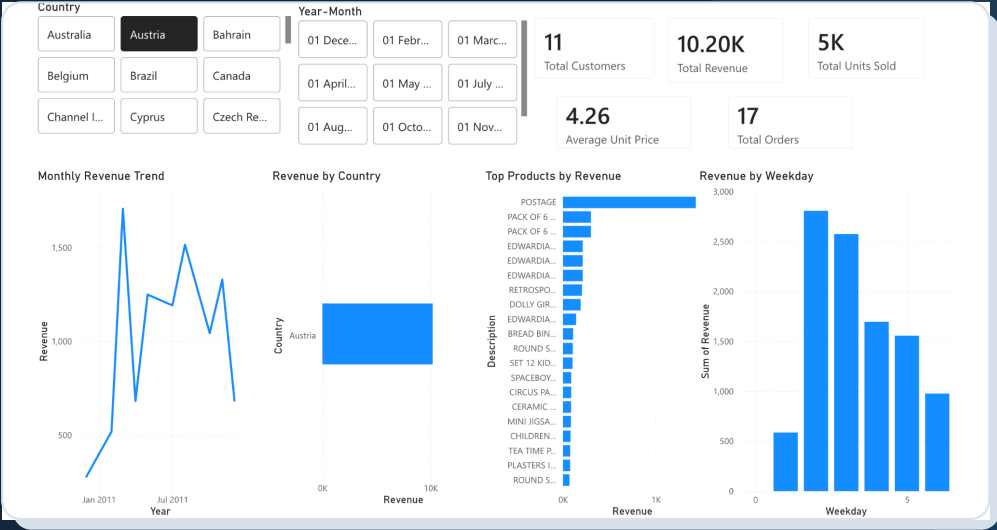
Databricks

PySpark

SQL

Delta Lake

Power BI

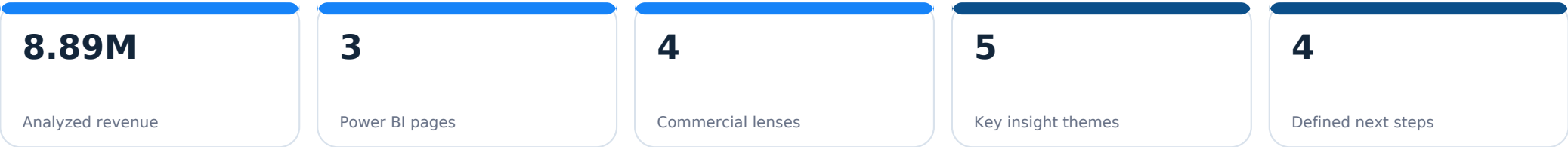


Valtteri Paivinen

Data and BI portfolio | 2026

Project at a glance

Business question, data scope and reporting outcomes



Business objective

- Track revenue performance and changes over time.
- Identify high-value customers and best-performing products.
- Compare market performance across countries and time periods.
- Translate commercial patterns into practical growth recommendations.

Dataset and deliverables

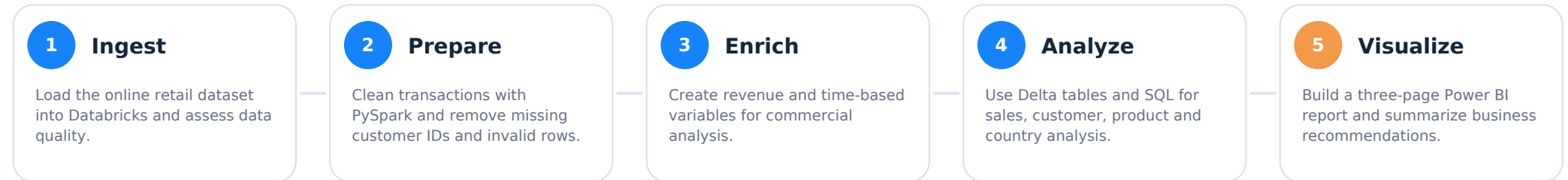
Public online retail transaction data containing invoice, product, quantity, date, unit price, customer and country-level fields.

Layer	Deliverable
Data	Cleaned, analysis-ready dataset
Analysis	Reusable SQL queries and segmented findings
Reporting	Interactive Power BI dashboard
Documentation	GitHub repository and overview PDF

Independent portfolio project created for skills demonstration; it does not represent paid client work.

From raw data to decision-ready reporting

A documented workflow combining data engineering, SQL analysis and BI storytelling



Technical approach

Databricks

Workspace for ingestion, transformation and analysis

PySpark

Repeatable cleaning, enrichment and feature creation

SQL + Delta

Segment analysis and persistent reporting tables

Power BI

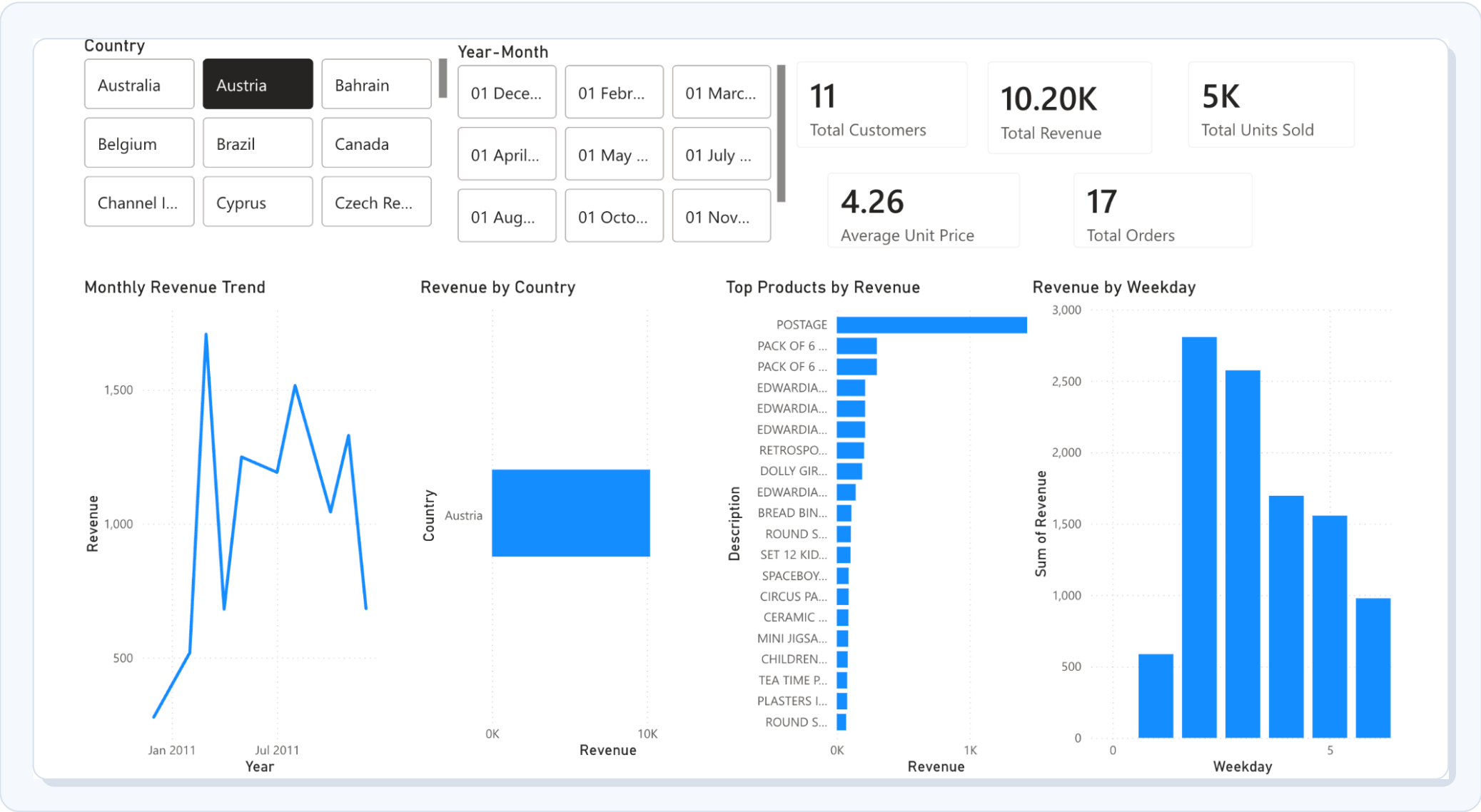
Interactive KPI reporting and business interpretation

Quality and interpretation

- Excluded transactions unsuitable for customer-level analysis.
- Standardized data types before revenue and time calculations.
- Created reusable revenue and calendar variables for reporting.
- Documented the analytical workflow and portfolio deliverables in GitHub.

Sales performance overview

How are revenue, orders and units changing across markets and time?



Customer and product performance

Who and what generate the most commercial value?

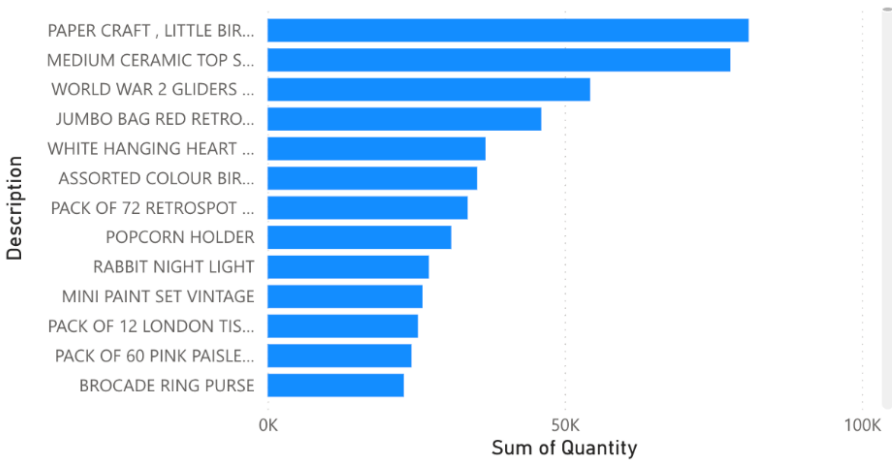
Top Revenue-Generating Customers

Customer ID	Total Revenue	Orders	Units sold
12346	77,183.60	1	74215
12415	124,914.53	21	77374
14156	117,210.08	55	57768
14646	280,206.02	73	196915
14911	143,711.17	201	80240
16029	80,850.84	63	40108
16446	168,472.50	2	80997
17450	194,390.79	46	69973
17511	91,062.38	31	64549
18102	259,657.30	60	64124
Total	1,537,659.21	553	806263

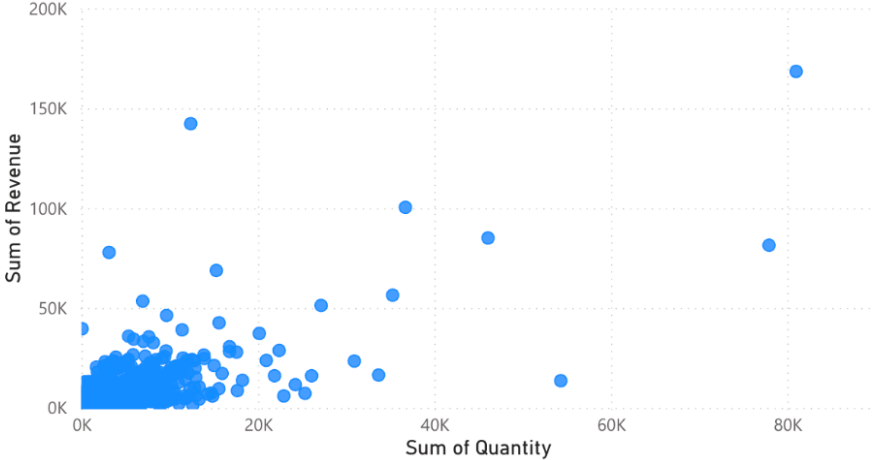
Revenue by Country and Year

Country	2010	2011	Total
Australia	965.35	137,488.46	138,453.81
Austria	277.20	9,921.48	10,198.68
Bahrain		548.40	548.40
Belgium	1,809.91	39,386.43	41,196.34
Brazil		1,143.60	1,143.60
Canada		3,666.38	3,666.38
Channel Islands	363.53	20,077.01	20,440.54
Cyprus	1,587.07	11,915.78	13,502.85
Czech Republic		826.74	826.74
Denmark	1,281.50	17,673.84	18,955.34
EIRE	8,813.88	256,448.58	265,262.46
European Community		1,300.25	1,300.25
Finland	892.80	21,653.28	22,546.08
France	9,616.31	199,336.00	208,952.31
Total	570,422.73	8,316,804.16	8,887,226.89

Best-Selling Products by Units Sold



Product Performance: Revenue vs Units Sold



Insights and recommendations

How can the findings support targeting, inventory and planning?

Insights & Recommendations

Key Insights

1. Revenue is strongly concentrated in a few key markets, especially the United Kingdom.
2. A small group of products generates a large share of total sales.
3. Some customers contribute significantly more revenue than others.
4. Monthly revenue trends show clear changes in sales performance over time.
5. Product-level analysis shows differences between high-volume and high-revenue products.

Business Recommendations

1. Focus marketing campaigns on high-value customers.
2. Prioritize best-selling products in inventory and promotion planning.
3. Develop country-specific strategies for the strongest markets.
4. Monitor monthly sales trends to identify seasonality and demand changes.
5. Use product and customer segmentation to improve targeting and sales performance.

Next Steps

1. Build customer segmentation using RFM analysis.
2. Add product cost data to analyze profitability.
3. Create sales forecasts for future months.
4. Connect sales data with marketing campaign data.

Key findings and recommended actions

Descriptive evidence translated into a practical decision framework

Key findings

- Revenue is concentrated in a small number of key markets, especially the United Kingdom.
- A limited group of products generates a substantial share of sales.
- Some customers contribute significantly more revenue than others.
- Monthly revenue trends show meaningful changes in sales performance over time.
- Product analysis separates high-volume items from high-revenue items.

Business recommendations

- Focus marketing activity on high-value customer groups.
- Prioritize best-selling products in inventory and promotion planning.
- Develop country-specific strategies for the strongest markets.
- Monitor monthly trends to identify seasonality and demand changes.
- Use product and customer segmentation to improve targeting.

Next steps

- Build customer segmentation with RFM analysis.
- Add product cost data to evaluate profitability.
- Create future-month sales forecasts.
- Connect transaction data with marketing-campaign data.

What this project demonstrates

- Sales and revenue analytics
- PySpark data preparation
- SQL customer and product analysis
- Delta table workflow
- Power BI KPI reporting
- Commercial recommendation writing

Findings describe associations in the analyzed data and should not be interpreted as causal effects.

github.com/valtteripaivinen/online-retail-sales-analysis